Hands-on: Three Operator Cases

Run voltage support, service restoration and cyber response in Classroom Labs

120 minutes / one CPU notebook / three short exercises

IEEE ComSoc theme / Run, change, check and explain

IEEE ComSoc IEEE Communications Society | IEEE

Independent educational material / IEEE ComSoc theme adaptation

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Open one notebook for all three cases

Open Classroom Labs in Google Colab

Classroom_Labs.ipynb

- Open Classroom Labs using the source link below or the website's first Colab button.
- Save a copy in Drive so you can keep your parameter changes and outputs.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Connect a CPU and run the setup cell

- Connect a standard CPU runtime. Run "Set up Google Colab (CPU)" and wait for Ready.
- Setup retrieves the supporting data, saved weights and records. It installs the required packages.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Find cells by L label, not by cell number

- Use the section headings L0 through L12 to find the next task.
- Run a whole code cell with its play button or Shift+Enter. Later cells use earlier variables.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Run L0: choose the classroom route

Classroom Labs / L0 / classroom-002

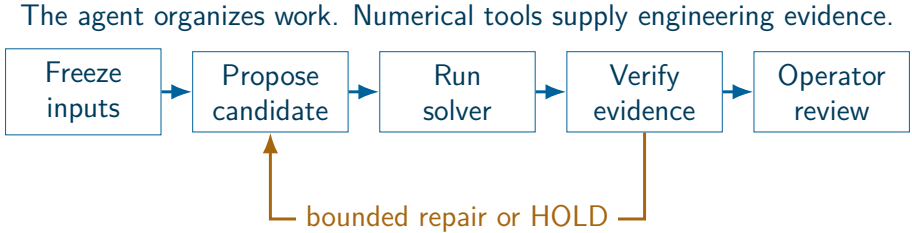
Classroom_Labs.ipynb L0 (classroom-002), source lines 11–14.

```
RUN_OPENDSS = False       # True: fresh full-model audit +
   separate AC MPC solve in L5
RUN_LOCAL_LLAMA = False   # True: one optional local call in L11
ROOT = Path.cwd()
assert (ROOT / 'data/lindistflow3_network.json').exists(), 'Start
   Jupyter in the tutorial folder'
```

- Keep both switches False. The class uses fresh linear-model calculations and saved model replies.
- If the data check fails, rerun the setup cell before continuing.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

Know which evidence you are reading



Teaching diagram of the model or check used at this notebook step.

- FRESH: calculated now. SAVED POLICY: old weights used for a new proposal.
- ARCHIVED ACTUAL LLM: a previously recorded reply, with its original input and timing.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

Run L1: does old evidence still match?

Classroom Labs / L1 / classroom-004

Executed notebook output

L1 · A tool contract is more than a prompt

classroom-004 · output 0 (zero-based) · recorded execution 3

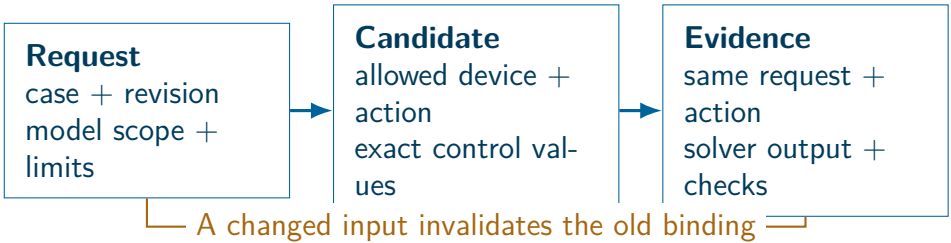
	test	binding_pass
0	same request + same command	True
1	changed load; old evidence	False
2	changed command; old evidence	False

Actual executed notebook output; source cell classroom-004. Fresh numerical calculation unless explicitly marked archived.

- Run the cell, then read binding_pass for all three rows.
- Expect True, False, False: changing the load or command invalidates this evidence binding.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

Use the same cycle in every case



Teaching diagram of the model or check used at this notebook step.

- Predict to run the default to read the result to change one input to rerun.
- Finish by naming the permitted next step and the evidence that supports it.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

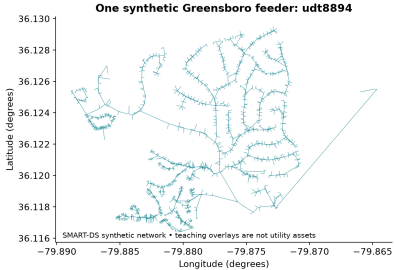
Keep a short experiment record

- Record the cell label and parameter value you changed.
- Record one result, its model scope and your next decision: continue, recheck or HOLD.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Voltage: start with L2

Classroom Labs / L2 / classroom-007



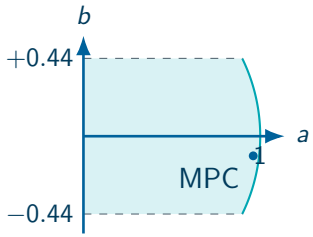
Synthetic SMART-DS feeder model used by the voltage example; not an operational utility circuit.

- Study one synthetic feeder. The command controls fleet active and reactive participation.
- Your task: check feasibility first, then compare the current-step objective.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Read the control and limits

Classroom Labs / L2 / classroom-007



$$p_i = aS_i, \qquad q_i = bS_i$$

$$0 \le a \le 1, \quad |b| \le 0.44, \quad a^2 + b^2 \le 1$$

$b < 0$: reactive absorption

Inverter capability is one check.

Teaching diagram of the model or check used at this notebook step.

- a sets active-power participation; b sets reactive-power participation.
- The notebook checks the inverter bounds, modeled voltage range and allowed current.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Run L2 with the default inputs

Classroom Labs / L2 / classroom-007

Classroom_Labs.ipynb L2 (classroom-007), source lines 1–4.

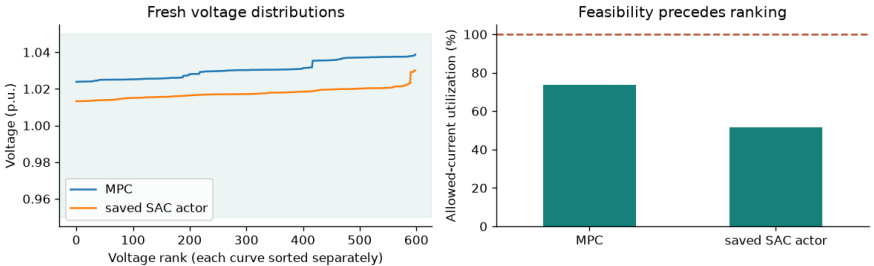
```
LOAD_SCALE = 1.0
RATING_FRACTION = 0.80
started = time.perf_counter()
voltage = lab.voltage_lab(LOAD_SCALE, RATING_FRACTION)
```

- Run the whole L2 cell. It solves MPC, evaluates the saved SAC proposal and checks both.
- Find feasible, current_step_J and max_allowed_current_pct in the output.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Read your voltage plots

Classroom Labs / L2 / classroom-007



Actual executed notebook output; source cell classroom-007. Fresh numerical calculation unless explicitly marked archived.

- Left: compare the sorted modeled voltage distributions with the allowed band.
- Right: compare current use with the 100% allowed-current line.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Read feasibility before cost

Classroom Labs / L2 / classroom-007

Executed notebook output

L2 · Model-based planning and model-free action proposals

classroom-007 · output 1 (zero-based) · recorded execution 4

	method	feasible	current_ step_J	a	b	vmin	vmax	max_allowed_ current_pct	failures
0	MPC	True	0.031104	0.960307	-0.111114	1.023874	1.038709	73.619901	none
1	saved SAC actor	True	0.436428	0.340066	0.014024	1.013243	1.030000	51.679336	none

Actual executed notebook output; source cell classroom-007. Fresh numerical calculation unless explicitly marked archived.

- Default MPC: PASS, current use 73.62%, current-step J about 0.03110.
- Default saved SAC: PASS, current use 51.68%, J about 0.43643.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Exercise 1: tighten the current allowance

Classroom Labs / L2 / classroom-007

```
RATING_FRACTION = 0.75
```

- In L2, change only RATING_FRACTION from 0.80 to 0.75. Predict, then rerun L2.
- For each method, report feasible, allowed-current use and current_step_J.
- Does passing this check establish full MV/LV feasibility? Explain your answer.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Classroom Labs / L2 / classroom-007

Both candidates still pass

- At 0.75, both candidates still pass: MPC about 78.53%, SAC about 54.85%.
- A stricter allowance raises utilization here, but it does not force a failure.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Reset L2 before the recorded-choice lesson

Classroom Labs / L2 / classroom-007

Classroom_Labs.ipynb L2 (classroom-007), source lines 1–2.

```
LOAD_SCALE = 1.0
RATING_FRACTION = 0.80
```

- Restore the default values and rerun L2 to keep your baseline easy to reproduce.
- L3–L5 use the original test_01 context. They do not inherit your edited L2 experiment.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

L3 starts a separate recorded context

Classroom Labs / L3 / classroom-010

Classroom_Labs.ipynb L3 (classroom-010), source lines 1–2.

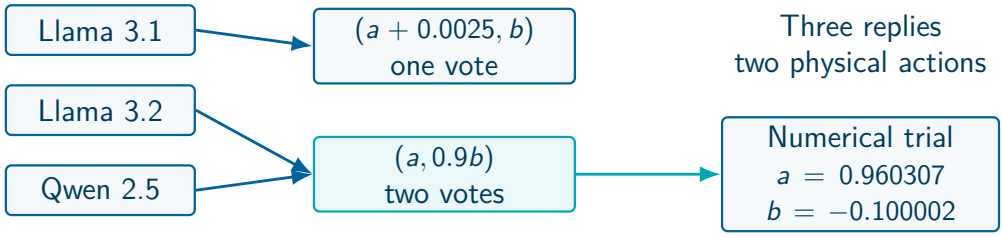
```
pool = lab.saved_pool('test_01')
display(pd.DataFrame(pool['responses'])[['model_id', 'operation_id
    ', 'reason', 'seconds', 'source']])
```

- Run L3. Read the original context ID, model operations and recorded seconds.
- The displayed reasons were recorded earlier. Opening them makes no new model call.

Source: Classroom_Labs.ipynb, L3; cell classroom-010; archived test_01 context.

What did the language models choose?

Classroom Labs / L3 / classroom-010



Teaching diagram of the model or check used at this notebook step.

- Different wording can request the same tool and parameters.
- The notebook groups identical physical trials; an explanation does not prove feasibility.

Source: Classroom_Labs.ipynb, L3; cell classroom-010; archived test_01 context.

Run L4: check the recorded choice again

Classroom Labs / L4 / classroom-013

Classroom_Labs.ipynb L4 (classroom-013), source lines 5–9.

```
started = time.perf_counter()
selected_trials = lab.trial_saved_selection(pool)
timings['L4 new verification of archived choices'] = time.
   perf_counter() - started
display(selected_trials['table'])
assert pool['frozen'] and pool['training_updates'] == 24
```

- Run L4 after L3. Python performs a fresh numerical check on the original selected candidate.
- Compare majority vote with frozen LinUCB on the same original context.

Source: Classroom_Labs.ipynb, L4; cell classroom-013; fresh checks of archived choices.

Separate the old choice from the new check

Classroom Labs / L4 / classroom-013

Executed notebook output

L4 · Learned response selection is a trial budget decision

classroom-013 · output 1 (zero-based) · recorded execution 6

	selector	operation	new_check_passes	current_step_J	verification_s	failures
0	majority vote	RELAX_Q_10_PERCENT	True	0.031082	0.091243	none
1	frozen LinUCB	RELAX_Q_10_PERCENT	True	0.031082	0.134089	none

Actual executed notebook output; source cell classroom-013. Fresh numerical calculation unless explicitly marked archived.

- Both selectors choose the same Q-relaxation trial and pass the fresh MV check.
- The predicted reward is not a feasibility result; read new_check_passes.

Source: Classroom_Labs.ipynb, L4; cell classroom-013; fresh checks of archived choices.

Run L5: identify the model scope

Classroom Labs / L5 / classroom-016

Classroom_Labs.ipynb L5 (classroom-016), source lines 1–4.

```
started = time.perf_counter()
scope_audit = lab.backend_audit(pool, run_opendss=RUN_OPENDSS)
timings['L5 selected model scope checks'] = time.perf_counter() -
    started
display(scope_audit['table'])
```

- Leave RUN_OPENDSS=False and run L5. Read both the table and optional_mode message.
- This checks the archived test_01 MV-MPC incumbent, with OpenDSS disabled.

Source: Classroom_Labs.ipynb, L5; cell classroom-016; MV-only default model.

A PASS belongs to the model that produced it

Classroom Labs / L5 / classroom-016

Executed notebook output

L5 · Same physical action, different model coverage

classroom-016 · output 0 (zero-based) · recorded execution 7

	model	command	fresh	passes	vmin	vmax	current_pct	failures
0	lindistflow3	same archived MV-MPC first action	True	True	1.023874	1.038709	73.661582	none

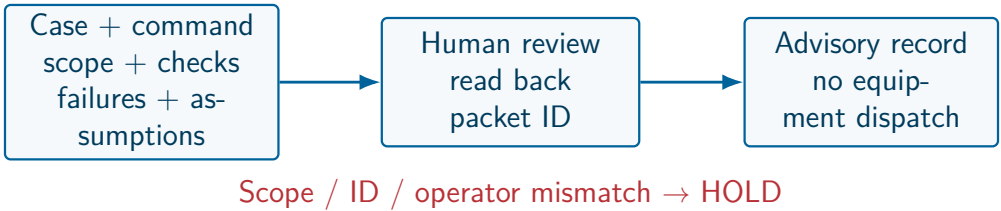
Actual executed notebook output; source cell classroom-016. Fresh numerical calculation unless explicitly marked archived.

- The default output checks MV phases and current. It does not cover all LV equipment.
- If the request needs full MV/LV coverage, request a broader-model check or HOLD.

Source: Classroom_Labs.ipynb, L5; cell classroom-016; MV-only default model.

Preview the handoff you will build at L12

Preview / L12 after all three cases



Teaching diagram of the model or check used at this notebook step.

- L12 uses selected_trials from L4, not the fresh L2 comparison.
- Finish L6–L11 first. Run L12 at the end of the workshop, in notebook order.

Source: Classroom_Labs.ipynb, L12; cell classroom-033; simulated readback only.

Preview: the two handoff outcomes

Preview / L12 / run at the end

Executed notebook output

L12 · Operator handoff: acknowledge the recommendation you actually checked

classroom-033 · output 0 (zero-based) · recorded execution 14

Contiguous columns: state through equipment_dispatch; original row order and row index.

	state	evidence_ matches	engineering_ pass	scope_ matches	scope	acknowled ged	equipment_ dispatch
0	HOLD	True	True	False	MV phases only	False	False
1	READY_FOR_ READBACK	True	True	True	MV phases only	True	False

Actual executed notebook output; source cell classroom-033. Fresh numerical calculation unless explicitly marked archived.

- Full MV/LV request: HOLD. Explicit MV teaching scope: simulated readback can succeed.
- A wrong recommendation ID is rejected. equipment_dispatch stays False.

Source: Classroom_Labs.ipynb, L12; cell classroom-033; simulated readback only.

Voltage checkpoint: state exactly what ran

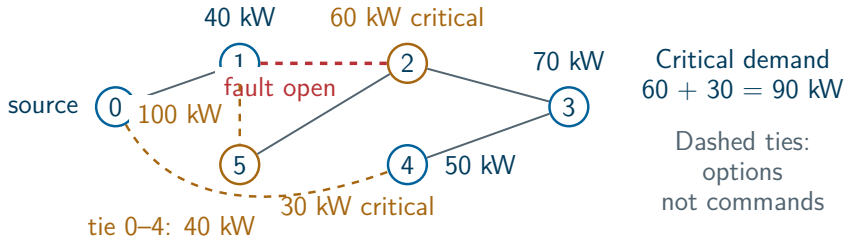
Classroom Labs / L5 / classroom-016

- L2: fresh MPC and saved-policy proposals. L3: old model replies. L4–L5: fresh checks of test_01.
- Report the input, candidate, model scope and outcome before proposing the next step.

Source: Classroom_Labs.ipynb, L5; cell classroom-016; MV-only default model.

Restoration: open L6

Classroom Labs / L6 / classroom-018



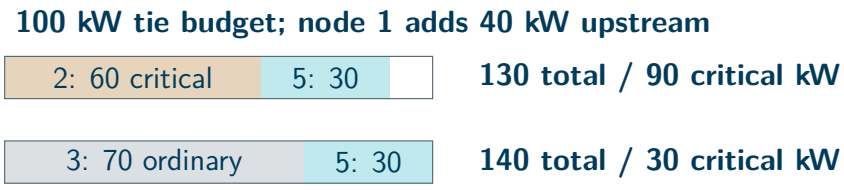
Teaching diagram of the model or check used at this notebook step.

- Use the six-node classroom graph. Keep the faulted branch unavailable.
- Your task: compare total restored load with service to critical loads.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Read the objective before choosing a plan

Classroom Labs / L6 / classroom-018



Teaching diagram of the model or check used at this notebook step.

- The priority plan weights two critical loads more heavily.
- The baseline maximizes total kW with the same graph, capacities and pickup rules.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Run L6: sweep the tie capacity

Classroom Labs / L6 / classroom-018

Classroom_Labs.ipynb L6 (classroom-018), source lines 1–5.

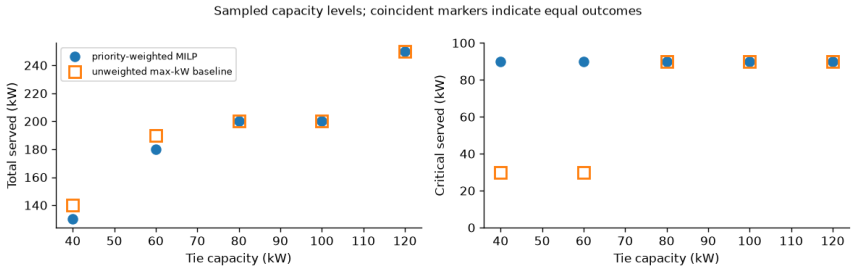
```
CAPACITIES_KW = (40., 60., 80., 100., 120.)
started = time.perf_counter()
restoration = lab.restoration_sweep(CAPACITIES_KW)
timings['L6 fresh MILP sweep + independent enumeration'] = time.
   perf_counter() - started
display(restoration['table'].drop(columns='pickup'))
```

- Run the whole L6 cell. It solves each capacity and checks the priority MILP by enumeration.
- Read served_kw, critical_kw and MILP_matches_exhaustive_check.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Read the two service plots together

Classroom Labs / L6 / classroom-018



Actual executed notebook output; source cell classroom-018. Fresh numerical calculation unless explicitly marked archived.

- Left: total served kW. Right: critical served kW.
- Coincident markers mean equal outcomes at that sampled capacity.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Explain the 40 kW tradeoff

Classroom Labs / L6 / classroom-018

Executed notebook output

L6 · Restoration: the objective expresses a service policy

classroom-018 · output 0 (zero-based) · recorded execution 8

Actual output header and rows 0–1 · tie_capacity_kw = 40.0

	tie_capacity_kw	method	served_kw	critical_kw	weighted_service	MILP_matches_exhaustive_check
0	40.0	priority-weighted MILP	130.0	90.0	940.0	True
1	40.0	unweighted max-kW baseline	140.0	30.0	410.0	True

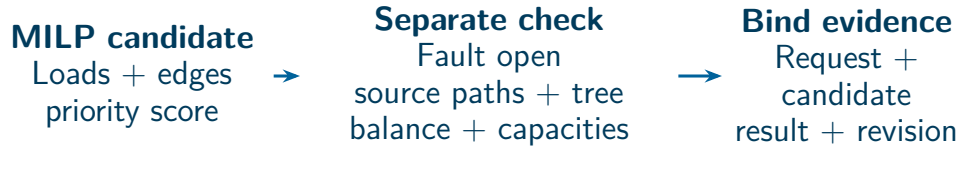
Actual executed notebook output; source cell classroom-018. Fresh numerical calculation unless explicitly marked archived.

- Priority objective: 130 kW total, including 90 kW critical.
- Maximum-total-kW objective: 140 kW total, including only 30 kW critical.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

How Python checks the plan

Classroom Labs / L6 / classroom-018



Teaching diagram of the model or check used at this notebook step.

- Check available source paths, capacities, balance and the faulted branch.
- Then compare the MILP objective with the independent exhaustive check.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Exercise 2: add more tie capacity

Classroom Labs / L6 / classroom-018

```
CAPACITIES_KW = (40., 60., 80., 100., 120., 180.)
```

- Append 180. to CAPACITIES_KW, keeping 40. as the first entry. Predict, then rerun L6.
- Compare total and critical service at 40 and 180 kW.
- At which of these settings do the two objectives agree?

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Debrief: enough capacity removes this tradeoff

Classroom Labs / L6 / classroom-018

250 kW total / 90 kW critical

- At 180 kW, both methods serve 250 kW total and 90 kW critical.
- The priority policy matters under constrained capacity; it does not improve every setting.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Reset the sweep; then run L7

Classroom Labs / L7 / classroom-021

Classroom_Labs.ipynb L7 (classroom-021), source lines 1–3.

```
rest_record = restoration['records'][0]['weighted']
rest_request = {'task': 'restoration', 'capacity_kw':
   CAPACITIES_KW[0],
                'critical_nodes': [2, 5], 'model': '
   six_node_active_transport', 'revision': 1}
```

- Restore the original CAPACITIES_KW list and rerun L6 before L7.
- L7 binds the first 40 kW plan, then changes the request to 20 kW.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

Read the changed-evidence test

Classroom Labs / L7 / classroom-021

Executed notebook output

L7 · Restoration under a changed contract

classroom-021 · output 0 (zero-based) · recorded execution 9

```
Original contract: True
Changed rating with old evidence: False
Required response: HOLD old recommendation; request a fresh constrained solve.
```

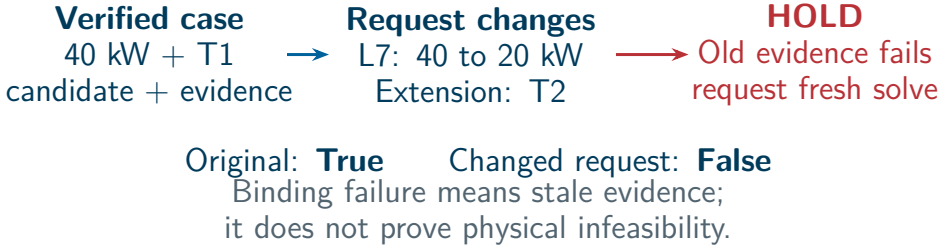
Actual executed notebook output; source cell classroom-021. Fresh numerical calculation unless explicitly marked archived.

- Original contract: True. Changed rating with old evidence: False.
- The stated next step is HOLD the old recommendation and request a fresh solve.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

A failed binding asks for a new check

Classroom Labs / L7 / classroom-021



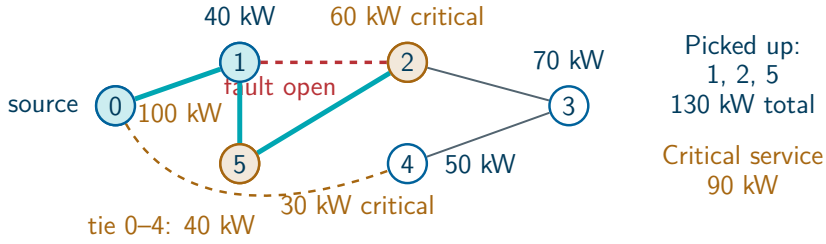
Teaching diagram of the model or check used at this notebook step.

- "The old evidence is stale" and "the new case is infeasible" are different claims.
- Update the request, solve the changed case, verify it and bind the new evidence.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

The agent must preserve the service policy

Classroom Labs / L6 / classroom-018



Teaching diagram of the model or check used at this notebook step.

- The operator specifies the objective and constraints; the MILP returns the pickup plan.
- The agent can request the right solve and explain its tradeoff using the returned evidence.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

State what the model leaves out

Classroom Labs / L6 / classroom-018

- This active-power graph does not certify AC voltage, reactive-power feasibility or protection.
- Its energized branches do not define a field switching sequence.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Restoration checkpoint: name the next action

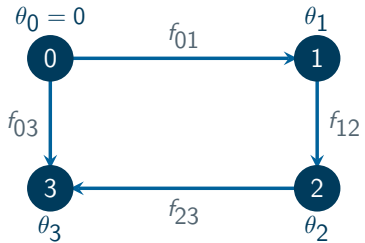
Classroom Labs / L7 / classroom-021

- For the 40 kW request, explain the 130/90 versus 140/30 service tradeoff.
- For a changed request, retain HOLD until current evidence is available.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

Cyber: open L8

Classroom Labs / L8 / classroom-023



Teaching diagram of the model or check used at this notebook step.

$x = (\theta_1, \theta_2, \theta_3)^\mathsf{T}$

4 flows + 3 injections + 1 angle

8 readings; $\mathrm{rank}(H) = 3$

Residual degrees: $8 - 3 = 5$

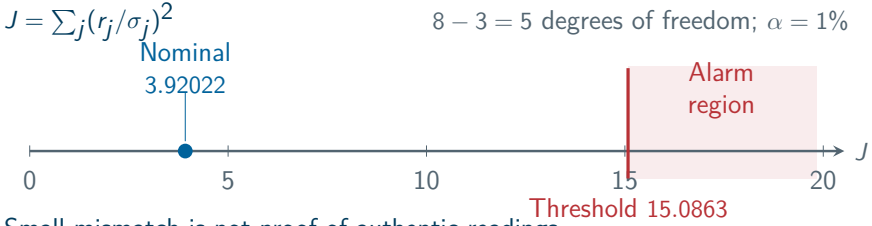
Fixed linear DC model

- Use the separate four-bus DC model and its supplied measurement data.
- Your task: distinguish a residual alarm from evidence about the true state.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Read residual and state error separately

Classroom Labs / L8 / classroom-023



Teaching diagram of the model or check used at this notebook step.

- J measures consistency between readings and the fitted model.
- State error compares the estimate with known synthetic truth; an operator does not get that truth.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Run L8: compare two attack patterns

Classroom Labs / L8 / classroom-023

Classroom_Labs.ipynb L8 (classroom-023), source lines 1–5.

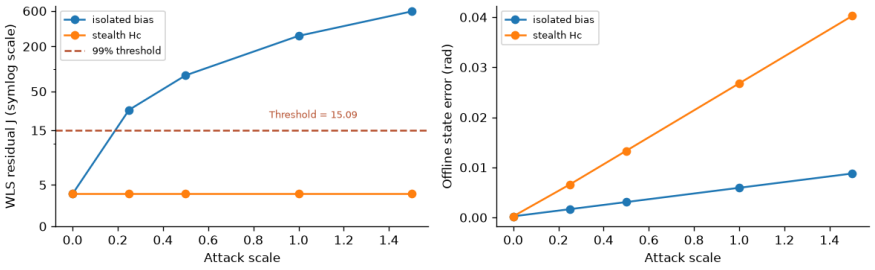
```
ATTACK_SCALES = (0., .25, .5, 1., 1.5)
started = time.perf_counter()
cyber = lab.cyber_sweep(ATTACK_SCALES)
timings['L8 fresh WLS attack sweep'] = time.perf_counter() -
   started
display(cyber['table'])
```

- Run the whole L8 cell with its default scale list.
- Compare isolated bias with coordinated Hc in both the table and plots.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Read your two plots together

Classroom Labs / L8 / classroom-023



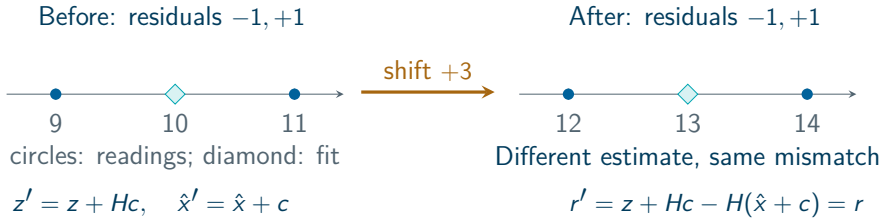
Actual executed notebook output; source cell classroom-023. Fresh numerical calculation unless explicitly marked archived.

- Left: residual J and the alarm threshold. Right: offline state error.
- Follow the coordinated curve across both panels before interpreting a quiet alarm.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

At scale 1, what does the alarm miss?

Classroom Labs / L8 / classroom-023



Teaching diagram of the model or check used at this notebook step.

- Isolated bias: J about 277.50, so the residual alarm is True.
- Coordinated Hc: J about 3.92, no alarm, but offline state error is about 0.02678 rad.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Exercise 3: add a small isolated bias

Classroom Labs / L8 / classroom-023

```
ATTACK_SCALES = (0., .10, .25, .5, 1., 1.5)
```

- Add .10 to ATTACK_SCALES. Predict, then rerun L8.
- At scale 0.10, read J, the alarm flag and the offline state error.
- Does "no alarm" mean the reading is correct? Use the two panels to explain.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Debrief: a quiet alarm is not proof of truth

Classroom Labs / L8 / classroom-023

$10.14 < 15.09$ to no residual alarm

- At scale 0.10, the isolated-bias residual is about 10.14, below the 15.09 threshold.
- A threshold can miss a small inconsistency. Coordinated corruption creates a different limitation.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Reset the sweep before recovery

Classroom Labs / L8 / classroom-023

Classroom_Labs.ipynb L8 (classroom-023), source lines 1–3.

```
ATTACK_SCALES = (0., .25, .5, 1., 1.5)
started = time.perf_counter()
cyber = lab.cyber_sweep(ATTACK_SCALES)
```

- Restore the original ATTACK_SCALES list and rerun L8.
- Next, L9 uses a fixed trusted-recovery fixture, not the scale you edited.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L9 is a fixed recovery example

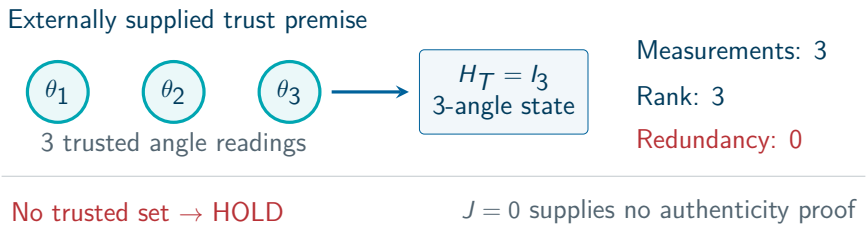
Classroom Labs / L9 / classroom-026

- The same supplied recovery case is checked with and without trusted replacement channels.
- Its control value does not track your L8 sweep-scale edit.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Where do the trusted readings come from?

Classroom Labs / L9 / classroom-026



Teaching diagram of the model or check used at this notebook step.

- The fixture assumes three independent angle readings are trustworthy.
- Neither the residual test nor a language-model explanation authenticates them.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Run L9: compare HOLD with PROPOSE

Classroom Labs / L9 / classroom-026

Classroom_Labs.ipynb L9 (classroom-026), source lines 1–2.

```
without_trust = lab.trusted_response(cyber, protected_available=
    False)
with_trust = lab.trusted_response(cyber, protected_available=True)
```

- Run the full L9 cell, including its assertions.
- Read status before comparing the returned injection or offline violation count.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Read the two decision states

Classroom Labs / L9 / classroom-026

Executed notebook output

L9 · Trusted state, constrained response, explicit HOLD

classroom-026 · output 0 (zero-based) · recorded execution 11

	condition	status	u_pu	offline_actual_violations
0	No trusted replacement channels	HOLD	NaN	NaN
1	Three trusted channels assumed	PROPOSE	0.121694	0.0

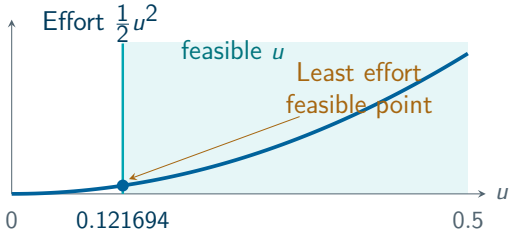
Actual executed notebook output; source cell classroom-026. Fresh numerical calculation unless explicitly marked archived.

- No trusted replacement channels: HOLD, with no proposed injection.
- Trusted channels assumed: PROPOSE, u about 0.121694 p.u., zero offline toy-model violations.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Check the numerical correction

Classroom Labs / L9 / classroom-026



All four branches:

$|\hat{f}(u)| \leq \ell - 0.01$

$0 \leq u \leq 0.5$ p.u.

Fixed DC sensitivities

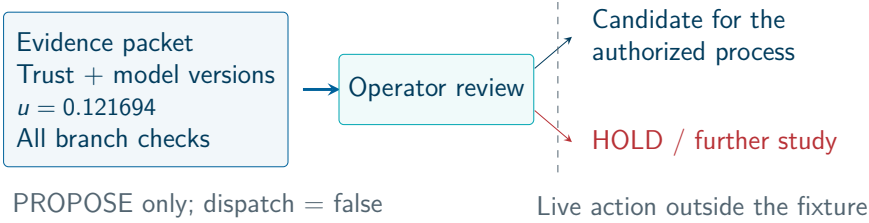
Teaching diagram of the model or check used at this notebook step.

- The QP minimizes effort while enforcing the declared branch limits and 0.01 p.u. margin.
- Predicted flow12 is 0.17000; offline actual is about 0.17046, below the 0.18000 rating.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Trace what caused the improvement

Classroom Labs / L9 / classroom-026



Teaching diagram of the model or check used at this notebook step.

- Independent trusted information supplies a usable estimate; the constrained solver supplies the action.
- The explanation must preserve those conditions and the remaining need for review.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Agent request, tool result, operator decision

Classroom Labs / L9 / classroom-026

- Agent role: request the allowed estimator/control tool and explain its returned conditions.
- Operator role: resolve the trust premise and review the bounded proposal through the authorized process.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Cyber checkpoint: state the condition behind PROPOSE

Classroom Labs / L9 / classroom-026

- Report the measurement-trust assumption, model, proposed u and failed or passed checks.
- If independent trusted information is unavailable, retain HOLD and name the missing evidence.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Run L10–L11 with the evidence labels visible

Classroom Labs / L10 / classroom-028

- L10 displays a separate archived five-stage study. It does not replay your edited L2, L6 or L8 inputs.
- Run L11 with RUN_LOCAL_LLAMA=False. The expected message says the new model call is skipped.

Source: Classroom_Labs.ipynb, L10–L11; cells classroom-028 and classroom-030.

Finish L12: practice the operator readback

Classroom Labs / L12 / classroom-033

Classroom_Labs.ipynb L12 (classroom-033), source lines 9–10.

```
wrong_id = lab.acknowledge(mv_scope, 'wrong-id', 'Student operator
    ')
readback = lab.acknowledge(mv_scope, mv_scope['recommendation_id
    '], 'Student operator')
```

- Run the full L12 cell now, after completing L0–L11 in notebook order.
- Check wrong-ID rejection, correct simulated acknowledgment and equipment_dispatch=False.

Source: Classroom_Labs.ipynb, L12; cell classroom-033; simulated readback only.

Save a reproducible workshop record

- Save your notebook copy with the parameter changes and outputs.
- Explain one case using: input to tool call to result to check to next decision.
- Keep fresh calculations, archived model replies and simulated readbacks clearly labeled.

Open the Full Reading Guide

Separate reading-guide PDF

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.